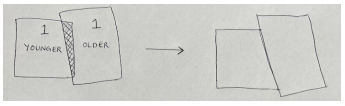
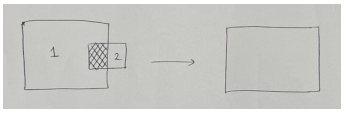
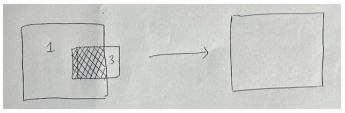
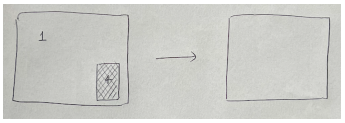
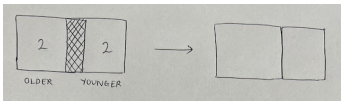
Decision Tree

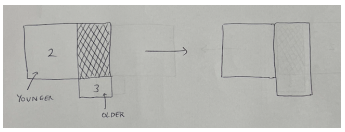
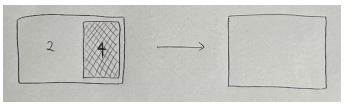
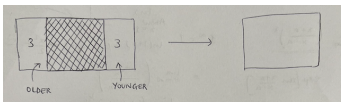
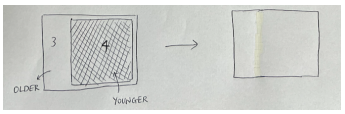
This is to make sure our decision tree covers all cases.

Intersect Area/Polygon Area (%)	100-80	80-50	50-20	20-0
Code	4	3	2	1

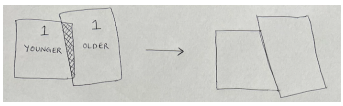
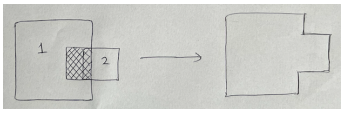
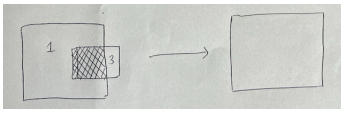
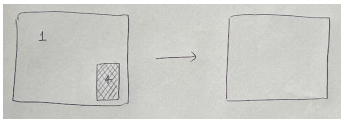
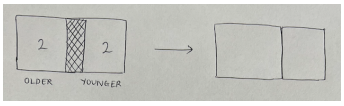
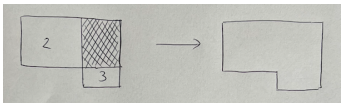
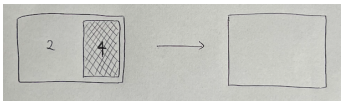
We work under the assumption that trees are being double counted.

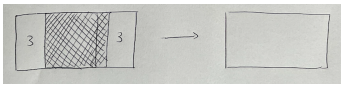
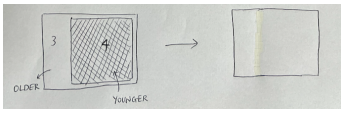
Harvey: Can confirm I've manually checked once instance where the trees were exactly double counted.

polyA	polyB	Overlap action	How to deal with tree data	Image
1	1	Leave older alone, cut from younger	Leave both tree data alone.	
1	2	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not - check if smaller one's trees are subset of bigger one)	
1	3	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not - check if smaller one's trees are subset of bigger one)	
1	4	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not- check if smaller one's trees are subset of bigger one).	
2	2	Leave older alone, cut from younger	Multiply tree data by corresponding %. (check that the older track knows about newer trees).	

2	3	Leave older alone, cut from younger	Discard younger polygon's tree data (check later if trees are double counted or not- check if smaller one's trees are subset of bigger one).	
2	4	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not).	
3	3	Discard younger polygon	Discard younger polygon's tree data (check later if trees are double counted or not- check if smaller one's trees are subset of bigger one).	
3	4	Discard younger polygon	Keep older polygon's tree data and discard younger's tree data. (Check if tree data is actually being double counted - do this by seeing if older one has newer trees associated with it.).	
4	4	Discard younger polygon	Keep older polygon's tree data and discard younger's tree data. (Check if tree data is actually being double counted - do this by seeing if older one has newer trees associated with it.).	

Redundant (**decided not to merge polygons**)

polyA	polyB	Overlap action	How to deal with tree data	Image
1	1	Leave older alone. Cut from younger	Leave both tree data alone.	
1	2	Merge polygons	Add % of smaller one's tree data to combined polygon.	
1	3	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not - check if smaller one's trees are subset of bigger one)	
1	4	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not- check if smaller one's trees are subset of bigger one).	
2	2	Leave older alone. Cut from younger	Multiply tree data by corresponding %. (check that the older track knows about newer trees).	
2	3	Merge polygons	Add % of smaller one's tree data to combined polygon. Check if trees are being double counted - if they are, tree data for smaller one should be ~ subset of bigger one).	
2	4	Discard smaller polygon	Discard smaller polygon's tree data (check later if trees are double counted or not).	

3	3	Merge polygons	Assuming double counting, merge tree data and then subtract overlapping proportion. (check later if trees are actually being double counted).	
3	4	Discard younger polygon	Keep older polygon's tree data and discard younger's tree data. (Check if tree data is actually being double counted - do this by seeing if older one has newer trees associated with it.).	
4	4	Discard younger polygon	Keep older polygon's tree data and discard younger's tree data. (Check if tree data is actually being double counted - do this by seeing if older one has newer trees associated with it.).	

Redundant (subset of the above)

% Overlap of intersection	100-80	80-20	20-0
Code	3	2	1

polyA	polyB	Type of intersection/action to do
1	1	Border overlap. Leave older alone. Remove intersection from younger
1	2	Merge the two and add the tree data.
1	3	A encompasses B and B is a small proportion of A. Keep A and discard B. Keep tree data from A and discard from B (we will need to check whether the tree data should be added instead).
2	2	Partial Overlap - Leave older alone. Remove intersection from younger and multiply its tree data by %overlap
2	3	A encompasses B but B is a significant proportion of A. Keep A and discard B. Keep tree data from A and discard from B (we will need to check whether the tree data should be added instead).
3	3	Matching - keep older polygon and keep tree data from older